

Harisree Purchase Assistant

Production Infrastructure & Deployment Documentation | Prepared by Anandu Krishna, HexaStack Solutions
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1. Overview

This document describes the live production setup for the Harisree Purchase Assistant (warehouse / stock & purchase management app). The system was migrated off Render (backend hosting) and Supabase (database) onto a self-hosted Windows Server, with a free, permanent public domain and HTTPS connection provided via Cloudflare Tunnel. The frontend remains hosted on Vercel.

2. Live URLs

Component	URL	Notes
Frontend (app)	https://purchase-assiastant.vercel.app	Flutter Web PWA, hosted on Vercel (free)
Backend API	https://api.harisreeagency.online	FastAPI backend via Cloudflare Tunnel
Backend health check	https://api.harisreeagency.online/health	Returns JSON status; use to verify backend is up
GitHub repository	https://github.com/ANANDU-2000/Purchase-Assiastant	Source code, both frontend & backend

3. Domain & DNS

Item	Detail
Domain	harisreeagency.online
Registrar	Namecheap
Purchase date	August 7, 2026
Renewal date	August 7, 2027 (auto-renew is ON)
First-year cost	~\$2 USD
DNS provider	Cloudflare (Free plan) — nameservers: coraline.ns.cloudflare.com, rene.ns.cloudflare.com
Cloudflare account	anandukrishna2999@gmail.com

4. Cloudflare Tunnel (how the backend is exposed)

Instead of opening router ports (which risks CGNAT issues and security exposure), the backend is exposed via a free, permanent **Cloudflare Tunnel**. A small background service (cloudflared) runs on the Windows Server and creates an outbound-only encrypted connection to Cloudflare, which then routes public traffic on

api.harisreeagency.online to the backend's local port.

Item	Detail
Tunnel name	harisree-backend
Connector	cloudflared, installed as a Windows service named "Cloudflared"
Public hostname route	api.harisreeagency.online → http://localhost:8000
Cloudflare Zero Trust plan	Free (no cost, card verification only, not charged)

5. Backend Server

Item	Detail
Machine	Windows 10, 2 CPU cores, 8 GB RAM (client's on-premise PC)
Remote access	AnyDesk ID: 797745244
Backend stack	FastAPI + Uvicorn (Python 3.13), running as a Windows service
Service manager	NSSM (Non-Sucking Service Manager)
NSSM path	C:\nssm\nssm-2.24\win64\nssm.exe
Service name	PurchaseAssistantAPI
Listening on	localhost:8000 (only reachable publicly via the Cloudflare Tunnel above)

Operational commands (run in Command Prompt as Administrator on the server):

Check backend status:

```
C:\nssm\nssm-2.24\win64\nssm.exe status PurchaseAssistantAPI
```

Start backend:

```
C:\nssm\nssm-2.24\win64\nssm.exe start PurchaseAssistantAPI
```

Stop backend:

```
C:\nssm\nssm-2.24\win64\nssm.exe stop PurchaseAssistantAPI
```

Restart backend (after code/ env changes):

```
C:\nssm\nssm-2.24\win64\nssm.exe restart PurchaseAssistantAPI
```

Check Cloudflare tunnel version / install:

```
cloudflared.exe --version
```

Check tunnel service (GUI):

Open services.msc → look for "Cloudflared" → Status should be "Running"

Verify backend is reachable publicly:

Open <https://api.harisreeagency.online/health> in any browser

6. Database

Item	Detail
Engine	PostgreSQL 18.4 (local install on the same Windows Server)

Database name	harisree_db
Host / port	localhost : 5432 (not exposed publicly — only the backend on the same machine connects to it)
Data origin	Restored from a Render production backup (pg_dump) during migration
Credentials	Stored in the backend's .env file on the server (not included in this document for security). Contact Anandu Krishna for access.

7. Frontend (Vercel)

Item	Detail
Framework	Flutter Web (built via a custom script: scripts/vercel-flutter-build.sh)
Hosting	Vercel — Hobby plan (free)
Live domain	purchase-assiastant.vercel.app
Key environment variable	API_BASE_URL = https://api.harisreeagency.online (set in Vercel → Settings → Environments)
Deploy trigger	Pushing to the "main" branch on GitHub auto-triggers a new Vercel build and deployment

Note: Because this is a Flutter Web build (not plain HTML/JS), changing API_BASE_URL in Vercel's dashboard alone is not enough — a fresh deployment (Redeploy, with build cache unchecked) is required afterward for the change to take effect, since the value is compiled into the app at build time.

8. GitHub Repository & Deployment Flow

Repository: github.com/ANANDU-2000/PurchaseAssiastant — contains both the Flutter frontend (flutter_app/) and the FastAPI backend (backend/).

- **Frontend changes:** push to GitHub main branch → Vercel automatically builds and deploys. No manual server steps needed.
- **Backend changes:** currently manual — pull the latest code on the Windows Server (git pull), then restart the PurchaseAssistantAPI service (see command above). Automated CI/CD for the backend is not yet set up (see Outstanding Tasks).

9. Cost Summary

Item	Cost	Frequency
Domain (harisreeagency.online, Namecheap)	~\$2 USD	Yearly (renews Aug 7, 2027)
Cloudflare DNS + Tunnel + Zero Trust	\$0 (Free plan)	N/A — no expiry, no upgrade required
Vercel hosting (frontend)	\$0 (Hobby plan)	N/A
Windows Server	Existing hardware	No recurring cloud cost

Total recurring cost	~\$2 USD / year	Domain renewal only
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10. Access & Ownership

Resource	Controlled by
GitHub repository	Anandu Krishna
Vercel project (frontend hosting)	Anandu Krishna
Namecheap domain account	Anandu Krishna
Cloudflare account (DNS + Tunnel + Zero Trust)	Anandu Krishna (anandukrishna2999@gmail.com)
Windows Server, backend service & database	Anandu Krishna / HexaStack Solutions — physically located at client premises, managed remotely via AnyDesk

11. Outstanding Tasks / Next Steps

- Set up automated database backups (not yet configured — currently no scheduled backup of harisree_db).
- Decommission the temporary ngrok tunnel now that Cloudflare Tunnel is live and verified.
- Set up GitHub Actions CI/CD for the backend, so a git push automatically deploys to the server (currently manual).
- Optionally add real AI provider API keys (OpenAI/Gemini/Groq) — the AI assistant currently runs in "stub / rules only" mode.

Document prepared for internal / team reference. For access credentials (server login, database password, admin panel login), contact Anandu Krishna directly rather than storing them in this document.